

```

Q1. #!/bin/bash
total=150
anonymous=$(echo "$total * 0.45" | bc)
smb=$(echo "$anonymous * 0.60" | bc)
spread=$(echo "$smb * 3 * 0.30" | bc)
total_compromised=$(echo "$anonymous + $spread" | bc)
risk=$(echo "$total_compromised / $total * 100" | bc -l)

```

```

echo "Initial systems compromised: $anonymous"
echo "SMB-susceptible systems: $smb"
echo "Expected second-stage spread: $spread"
echo "Estimated total compromised: $total_compromised"
echo "Percentage of network compromised: $risk%"

```

```

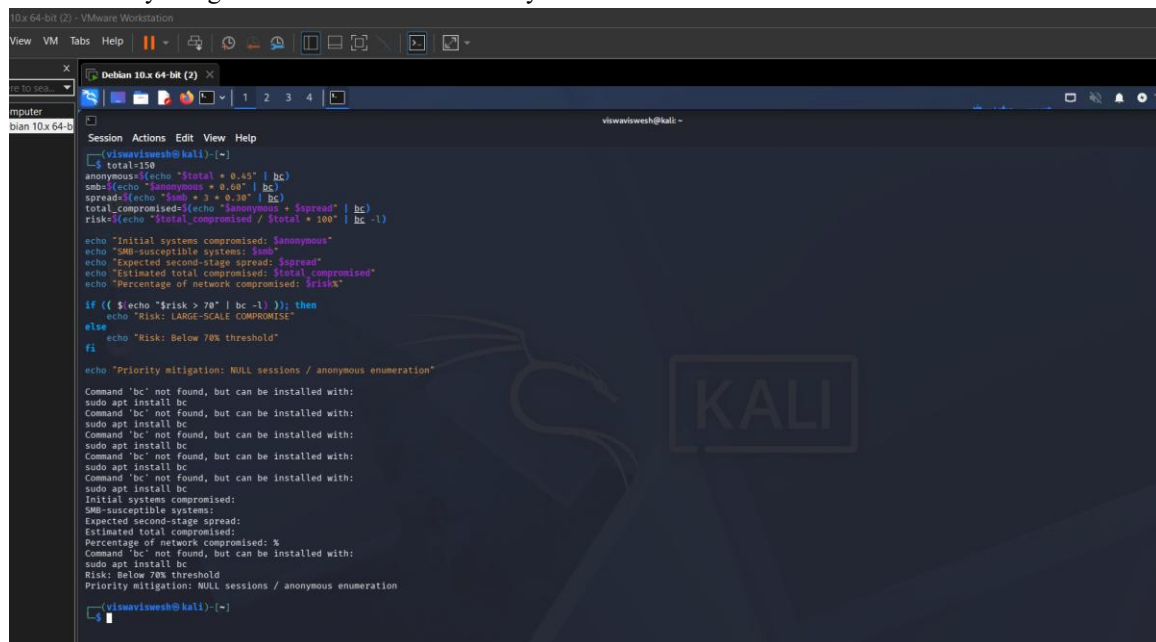
if (( $(echo "$risk > 70" | bc -l) )); then
    echo "Risk: LARGE-SCALE COMPROMISE"
else
    echo "Risk: Below 70% threshold"
fi

```

```

echo "Priority mitigation: NULL sessions / anonymous enumeration"

```



```

10x 64-bit (2) - VMware Workstation
View VM Tabs Help
Debian 10.x 64-bit (2)
vishwvishesh@kali: ~
Session Actions Edit View Help
vishwvishesh@kali:~$
total=150
anonymous=$(echo "$total * 0.45" | bc)
smb=$(echo "$anonymous * 0.60" | bc)
spread=$(echo "$smb * 3 * 0.30" | bc)
total_compromised=$(echo "$anonymous + $spread" | bc)
risk=$(echo "$total_compromised / $total * 100" | bc -l)

echo "Initial systems compromised: $anonymous"
echo "SMB-susceptible systems: $smb"
echo "Expected second-stage spread: $spread"
echo "Estimated total compromised: $total_compromised"
echo "Percentage of network compromised: $risk%"

if (( $(echo "$risk > 70" | bc -l) )); then
    echo "Risk: LARGE-SCALE COMPROMISE"
else
    echo "Risk: Below 70% threshold"
fi

echo "Priority mitigation: NULL sessions / anonymous enumeration"

Command 'bc' not found, but can be installed with:
sudo apt install bc
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sudo apt install bc
Command 'bc' not found, but can be installed with:
sudo apt install bc
Initial systems compromised:
SMB-susceptible systems:
Expected second-stage spread:
Estimated total compromised:
Percentage of network compromised: %
Command 'bc' not found, but can be installed with:
sudo apt install bc
Risk: Below 70% threshold
Priority mitigation: NULL sessions / anonymous enumeration
vishwvishesh@kali:~$

```

Q2. #!/bin/bash

```
total=120
```

```
rpc=70
```

```
vulnerable=$(echo "$rpc * 0.40" | bc)
```

```
successful=$(echo "$vulnerable * 0.50" | bc)
```

```
alerts=$(echo "$vulnerable * 0.25" | bc)
```

```
echo "RPC-exposed systems: $rpc"
```

```
echo "Vulnerable systems: $vulnerable"
```

```
echo "Expected successful compromises: $successful"
```

```
echo "Expected detection alerts: $alerts"
```

```
stealth=$(echo "$successful / $vulnerable * 100" | bc -l)
```

```
echo "Exploitation success rate: $stealth%"
```

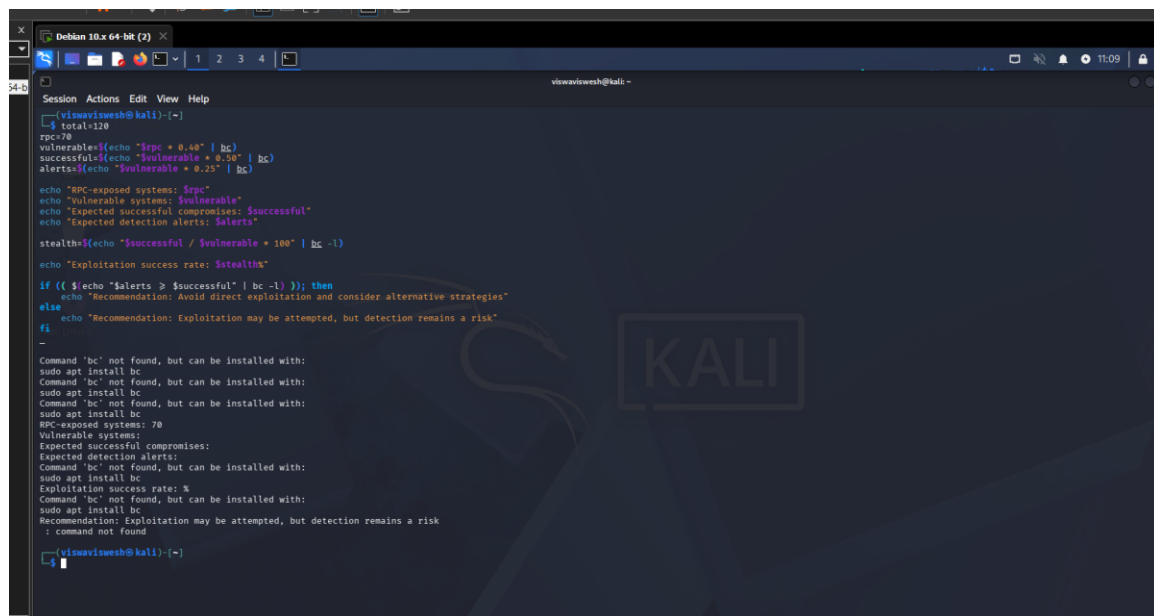
```
if (( $(echo "$alerts >= $successful" | bc -l) )); then
```

```
    echo "Recommendation: Avoid direct exploitation and consider alternative strategies"
```

```
else
```

```
    echo "Recommendation: Exploitation may be attempted, but detection remains a risk"
```

```
fi
```



```
view@viewsh@kali:~$ cat script.sh
#!/bin/bash

total=120
rpc=70
vulnerable=$(echo "$rpc * 0.40" | bc)
successful=$(echo "$vulnerable * 0.50" | bc)
alerts=$(echo "$vulnerable * 0.25" | bc)

echo "RPC-exposed systems: $rpc"
echo "Vulnerable systems: $vulnerable"
echo "Expected successful compromises: $successful"
echo "Expected detection alerts: $alerts"

stealth=$(echo "$successful / $vulnerable * 100" | bc -l)

echo "Exploitation success rate: $stealth%"

if (( $(echo "$alerts >= $successful" | bc -l) )); then
    echo "Recommendation: Avoid direct exploitation and consider alternative strategies"
else
    echo "Recommendation: Exploitation may be attempted, but detection remains a risk"
fi

view@viewsh@kali:~$ ./script.sh
RPC-exposed systems: 70
Vulnerable systems: 28
Expected successful compromises: 14
Expected detection alerts: 7
Exploitation success rate: 8%
Recommendation: Avoid direct exploitation and consider alternative strategies
```

Q3. #!/bin/bash

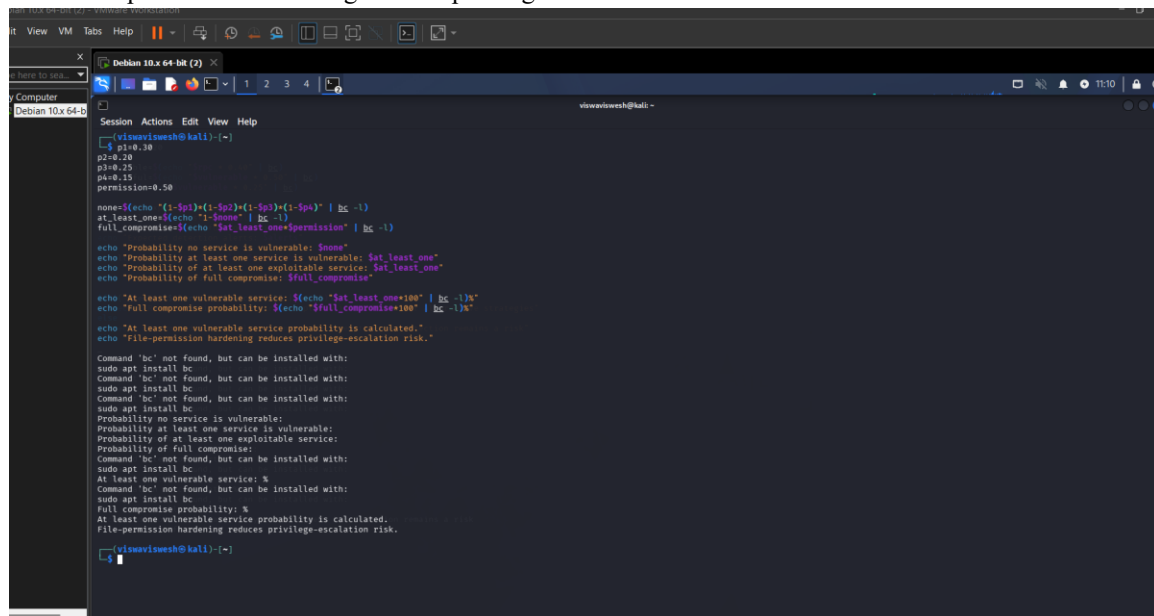
```
p1=0.30
p2=0.20
p3=0.25
p4=0.15
permission=0.50
```

```
none=$(echo "(1-$p1)*(1-$p2)*(1-$p3)*(1-$p4)" | bc -l)
at_least_one=$(echo "1-$none" | bc -l)
full_compromise=$(echo "$at_least_one*$permission" | bc -l)
```

```
echo "Probability no service is vulnerable: $none"
echo "Probability at least one service is vulnerable: $at_least_one"
echo "Probability of at least one exploitable service: $at_least_one"
echo "Probability of full compromise: $full_compromise"
```

```
echo "At least one vulnerable service: $(echo "$at_least_one*100" | bc -l)%"
echo "Full compromise probability: $(echo "$full_compromise*100" | bc -l)%"
```

```
echo "At least one vulnerable service probability is calculated."
echo "File-permission hardening reduces privilege-escalation risk."
```



```
view@kali: ~$ cat script.sh
#!/bin/bash

p1=0.30
p2=0.20
p3=0.25
p4=0.15
permission=0.50

none=$(echo "(1-$p1)*(1-$p2)*(1-$p3)*(1-$p4)" | bc -l)
at_least_one=$(echo "1-$none" | bc -l)
full_compromise=$(echo "$at_least_one*$permission" | bc -l)

echo "Probability no service is vulnerable: $none"
echo "Probability at least one service is vulnerable: $at_least_one"
echo "Probability of at least one exploitable service: $at_least_one"
echo "Probability of full compromise: $full_compromise"

echo "At least one vulnerable service: $(echo "$at_least_one*100" | bc -l)%"
echo "Full compromise probability: $(echo "$full_compromise*100" | bc -l)%"

echo "At least one vulnerable service probability is calculated."
echo "File-permission hardening reduces privilege-escalation risk."

Command 'bc' not found, but can be installed with:
sudo apt install bc
Command 'bc' not found, but can be installed with:
sudo apt install bc
Command 'bc' not found, but can be installed with:
sudo apt install bc
Probability no service is vulnerable:
Probability at least one service is vulnerable:
Probability of at least one exploitable service:
Probability of full compromise:
Command 'bc' not found, but can be installed with:
sudo apt install bc
At least one vulnerable service: %
Command 'bc' not found, but can be installed with:
sudo apt install bc
Full compromise probability: %
At least one vulnerable service probability is calculated.
File-permission hardening reduces privilege-escalation risk.

view@kali: ~$
```

Q4. #!/bin/bash

subnet1=0.40

subnet2=0.30

subnet3=0.20

root=0.70

none=\$(echo "(1-\$subnet1)*(1-\$subnet2)*(1-\$subnet3)" | bc -l)

unauthorized=\$(echo "1-\$none" | bc -l)

complete=\$(echo "\$unauthorized*\$root" | bc -l)

echo "Probability no subnet gains access: \$none"

echo "Probability at least one subnet gains access: \$unauthorized"

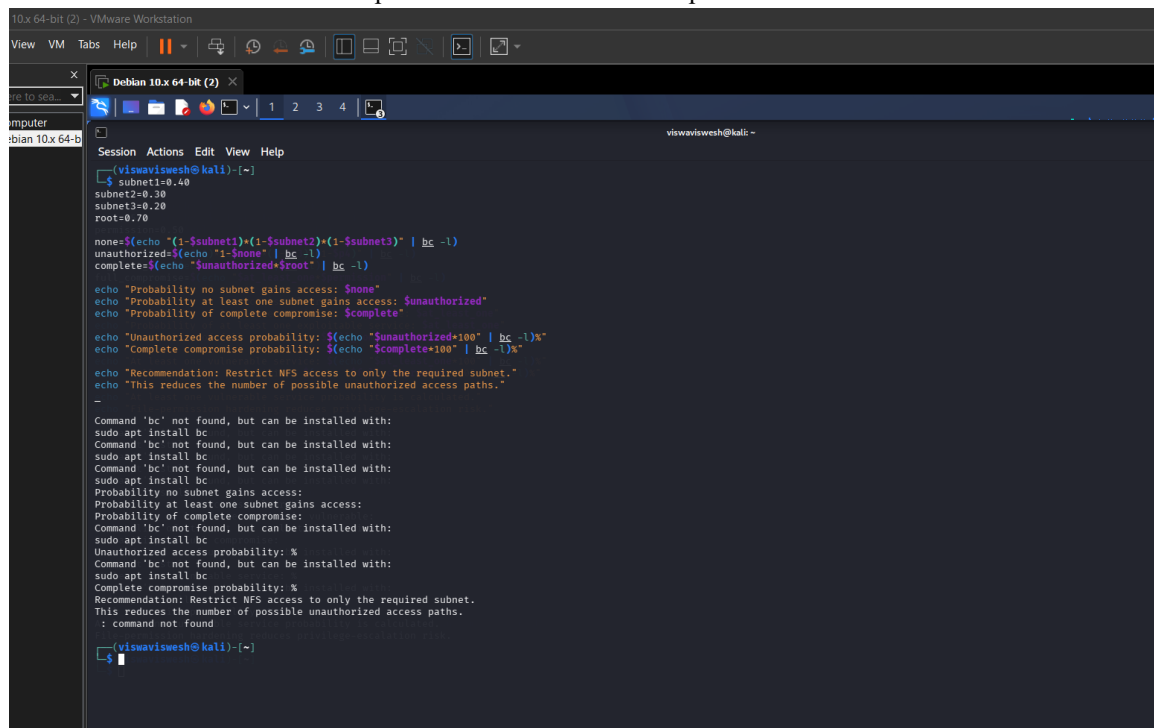
echo "Probability of complete compromise: \$complete"

echo "Unauthorized access probability: \$(echo "\$unauthorized*100" | bc -l)%"

echo "Complete compromise probability: \$(echo "\$complete*100" | bc -l)%"

echo "Recommendation: Restrict NFS access to only the required subnet."

echo "This reduces the number of possible unauthorized access paths."



```
10.x 64-bit (2) - VMware Workstation
View VM Tabs Help
Debian 10.x 64-bit (2)
computer
bian 10.x 64-b
Session Actions Edit View Help
(vismavishesh@kali)-[~]
└─$ subnet1=0.40
    subnet2=0.30
    subnet3=0.20
    root=0.70

    none=$(echo "(1-$subnet1)*(1-$subnet2)*(1-$subnet3)" | bc -l)
    unauthorized=$(echo "1-$none" | bc -l)
    complete=$(echo "$unauthorized*$root" | bc -l)

    echo "Probability no subnet gains access: $none"
    echo "Probability at least one subnet gains access: $unauthorized"
    echo "Probability of complete compromise: $complete"

    echo "Unauthorized access probability: $(echo "$unauthorized*100" | bc -l)%"
    echo "Complete compromise probability: $(echo "$complete*100" | bc -l)%"

    echo "Recommendation: Restrict NFS access to only the required subnet."
    echo "This reduces the number of possible unauthorized access paths."

-
Command 'bc' not found, but can be installed with:
sudo apt install bc
Command 'bc' not found, but can be installed with:
sudo apt install bc
Command 'bc' not found, but can be installed with:
sudo apt install bc
Probability no subnet gains access:
Probability at least one subnet gains access:
Probability of complete compromise:
Command 'bc' not found, but can be installed with:
sudo apt install bc
Unauthorized access probability: %
Command 'bc' not found, but can be installed with:
sudo apt install bc
Complete compromise probability: %
Recommendation: Restrict NFS access to only the required subnet.
This reduces the number of possible unauthorized access paths.
: command not found

(vismavishesh@kali)-[~]
└─$
```

Q5. #!/bin/bash

```
records=50000
```

```
sensitive_percent=0.30
```

```
rate=500
```

```
block_time=40
```

```
extracted=$(echo "$rate * $block_time" | bc)
```

```
percentage=$(echo "$extracted / $records * 100" | bc -l)
```

```
sensitive=$(echo "$extracted * $sensitive_percent" | bc)
```

```
echo "Total database records: $records"
```

```
echo "Extraction rate: $rate records/second"
```

```
echo "Attack duration: $block_time seconds"
```

```
echo "Records extracted: $extracted"
```

```
echo "Percentage of database compromised: $percentage%"
```

```
echo "Estimated sensitive records exposed: $sensitive"
```

```
if (( $(echo "$percentage >= 30" | bc -l) )); then
```

```
    echo "Severity: HIGH"
```

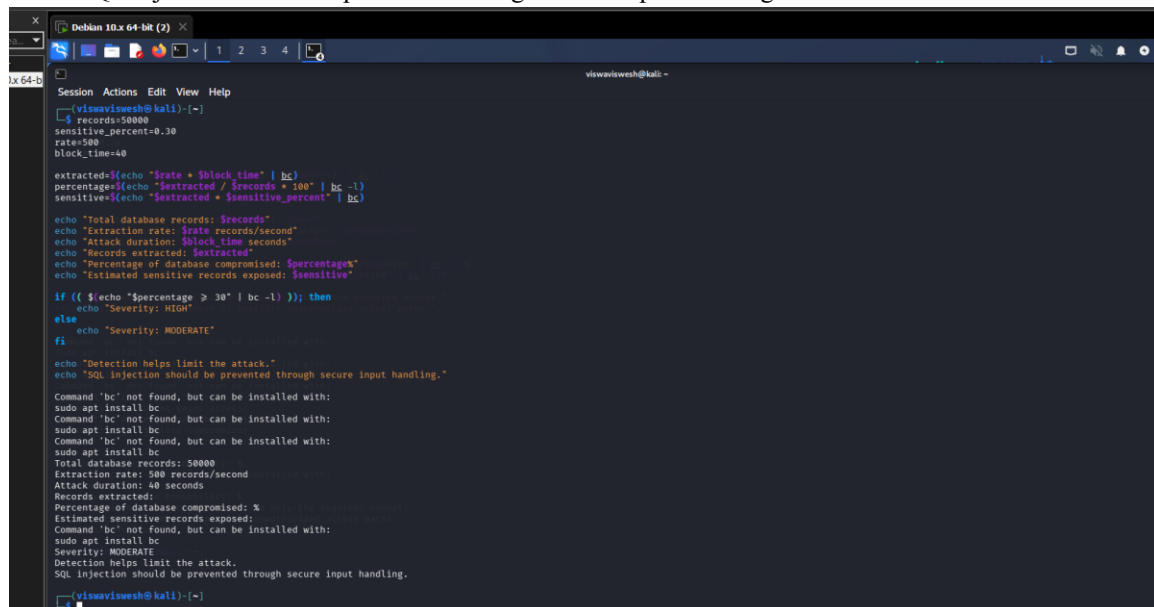
```
else
```

```
    echo "Severity: MODERATE"
```

```
fi
```

```
echo "Detection helps limit the attack."
```

```
echo "SQL injection should be prevented through secure input handling."
```



```
viswajish@kali:~$ cat script.sh
#!/bin/bash

records=50000
sensitive_percent=0.30
rate=500
block_time=40

extracted=$(echo "$rate * $block_time" | bc)
percentage=$(echo "$extracted / $records * 100" | bc -l)
sensitive=$(echo "$extracted * $sensitive_percent" | bc)

echo "Total database records: $records"
echo "Extraction rate: $rate records/second"
echo "Attack duration: $block_time seconds"
echo "Records extracted: $extracted"
echo "Percentage of database compromised: $percentage%"
echo "Estimated sensitive records exposed: $sensitive"

if (( $(echo "$percentage >= 30" | bc -l) )); then
    echo "Severity: HIGH"
else
    echo "Severity: MODERATE"
fi

echo "Detection helps limit the attack."
echo "SQL injection should be prevented through secure input handling."
```

```
viswajish@kali:~$ ./script.sh
Total database records: 50000
Extraction rate: 500 records/second
Attack duration: 40 seconds
Records extracted: 20000
Percentage of database compromised: 40%
Estimated sensitive records exposed: 8000
Severity: MODERATE
Detection helps limit the attack.
SQL injection should be prevented through secure input handling.
```